

Data Visualization & Storytelling

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About me

- Worked with LinkedIn as Insights Program Manager, Bangalore!
- Developing Design skills was the best part when I worked at Airbnb.
- Learned about been Scrappy at Mu Sigma

How I came to care about data visualization...

“ I was never good with drawing but I loved Colors”

“Storytelling and data visualization is Fun”

Why Data Visualization?

- Visualizations are data stories made by transforming the data quantitatively, defining relations, patterns and creating interactive images.
- Graphical or pictorial representation of data is Data visualization.
- It enables executives, business leaders, teachers, decision-makers to see analytics presented visually, so they can grasp difficult concepts or identify new patterns.
- Using tables or raw data, certain patterns, trends, and correlations might go unnoticed, and thus recognizing these patterns is easier with data visualization.
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Benefits of Data Visualization:

Considering the impact that data has in the growth of the business ,here are a few benefits:

- Helps to identify the latest trend to improve the product and increase the profits for business.
- Data visualizations make big and small data easier for the human brain to understand,leading to better analysis.
- Helps understanding the story-The human brain is not able to understand or even just imagine large amounts of numbers or text at once. It needs a visual representation to make sense of them and consequently translate raw data into tangible concepts.

**“Storytelling is more
important than plotting
complex graphs.”**

Remove to Improve: Less is More

Remove
to improve
the **map** edition

Agenda

Data Visualization & Storytelling

- External resources
- Principles of Data Visualization
- How to make good charts

External Resources for Data Visualization

Guidelines & Sources of Inspiration

Quick Reads

[5 Data Storytelling Tips for Creating More Persuasive Charts and Graphs](#)

[Ten Simple Rules for Better Figures](#) (favorite academic paper)

Blogs

[Stephen Few](#) (Perceptual Edge)

Darkhorse Analytics [blog](#)

Hans Rosling's 2006 [TED Talk](#) [storytelling]

Academic Tomes

[Edward Tufte](#) (ironically, worst website ever)

Principles of Data Visualization

1. Decide on your **message**
2. Aesthetics matter
3. Communicate clearly

1. Message

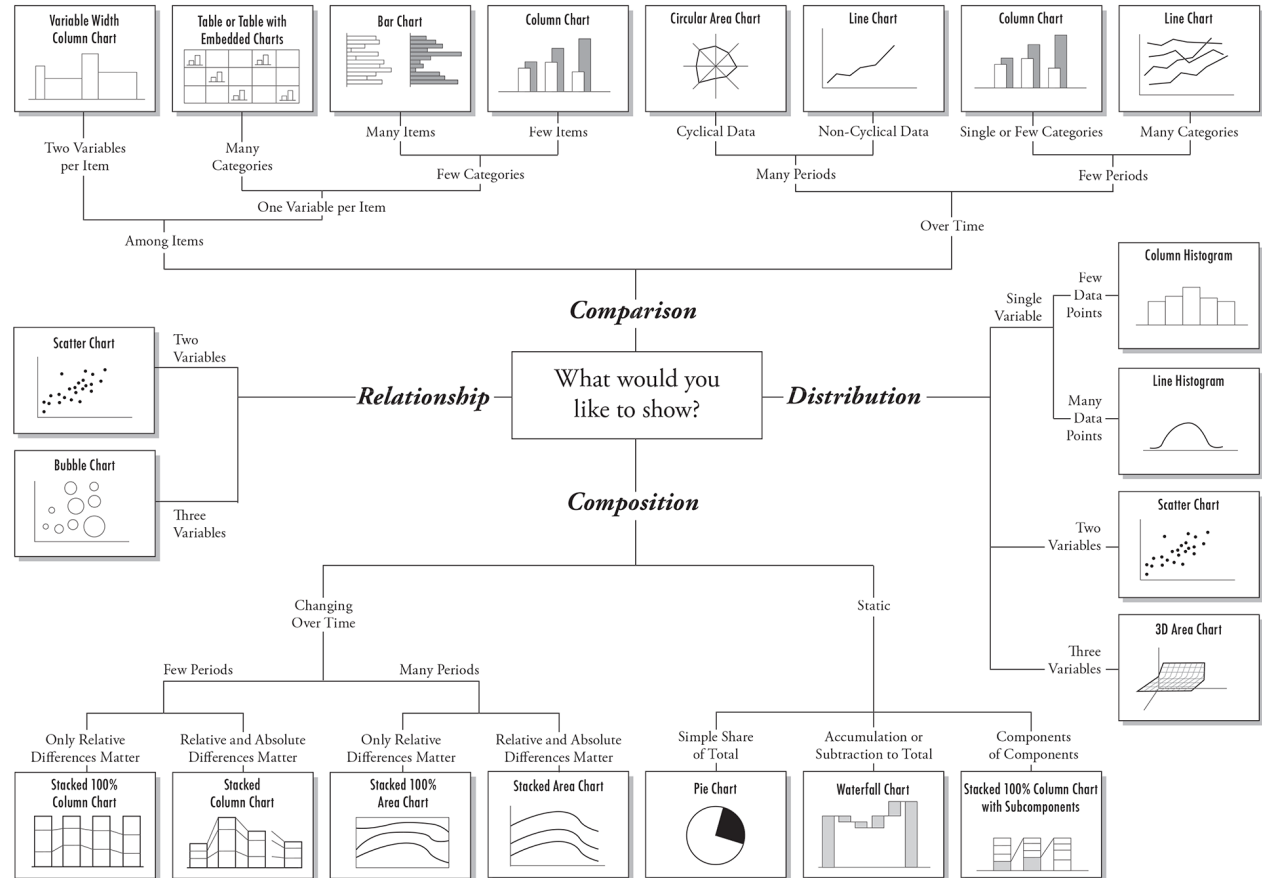
Make sure your figure supports your message.

“Signal must come first, even if your aim is primarily engagement”

1. Choose the right chart/graph for you message
2. Choose the right chart elements

Choosing the Right Chart

Chart Suggestions—A Thought-Starter



Choosing the Right Chart Element (Channel)

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Depth (3D position)	
Color luminance	
Color saturation	
Curvature	
Volume (3D size)	
Spatial region	
Color hue	
Motion	
Shape	

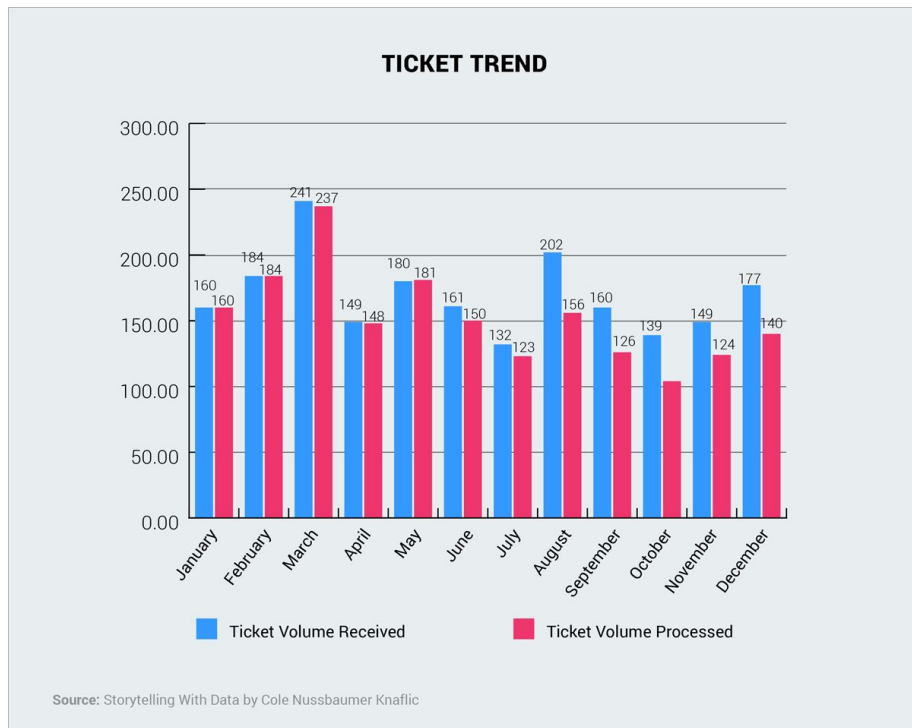
effective

Read more about [color](#)

effective

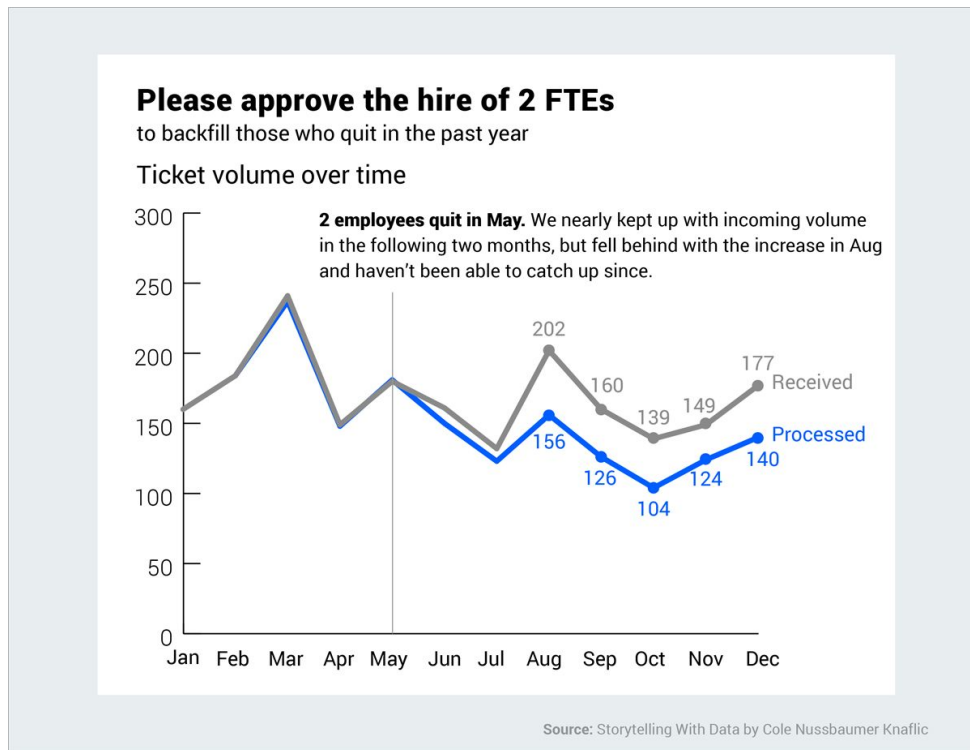
Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing difference



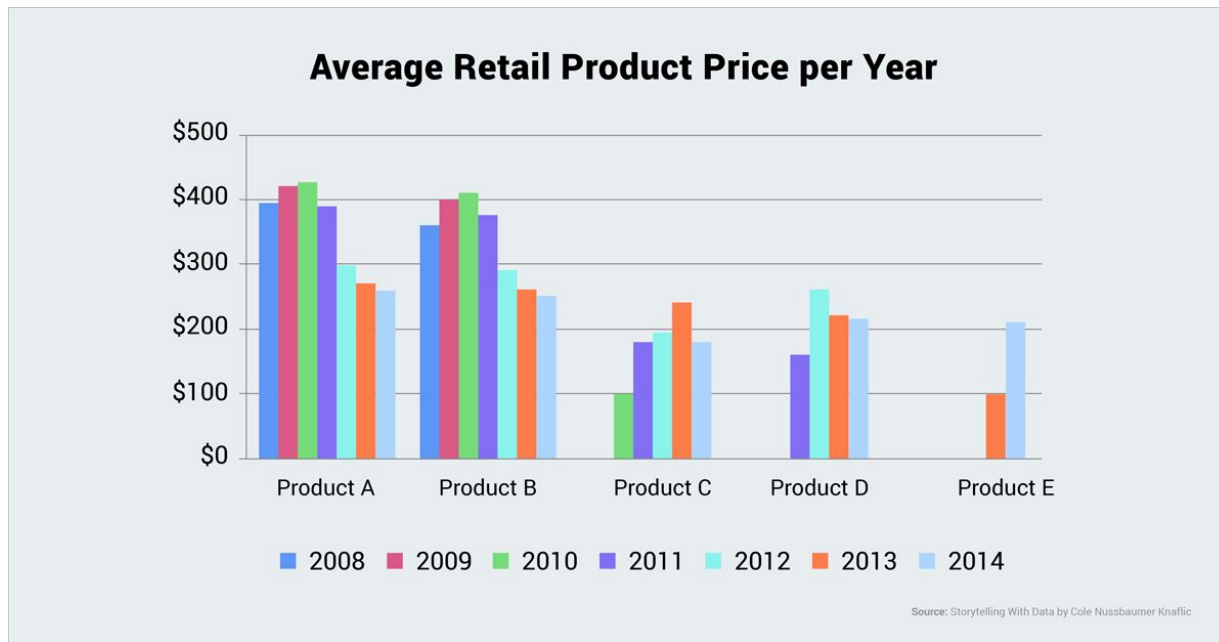
Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing difference



Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing (converging) similarity

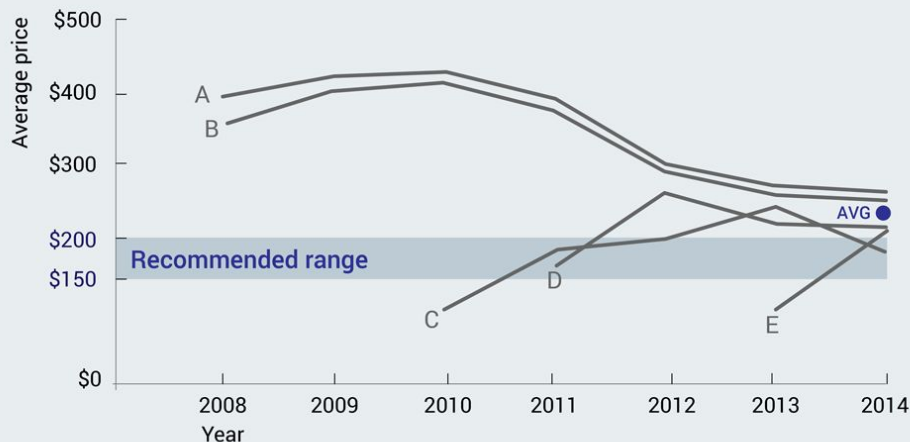


Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing (converging) similarity

To be competitive, we recommend introducing our product below the \$223 average price point in the **\$150-\$200 range**

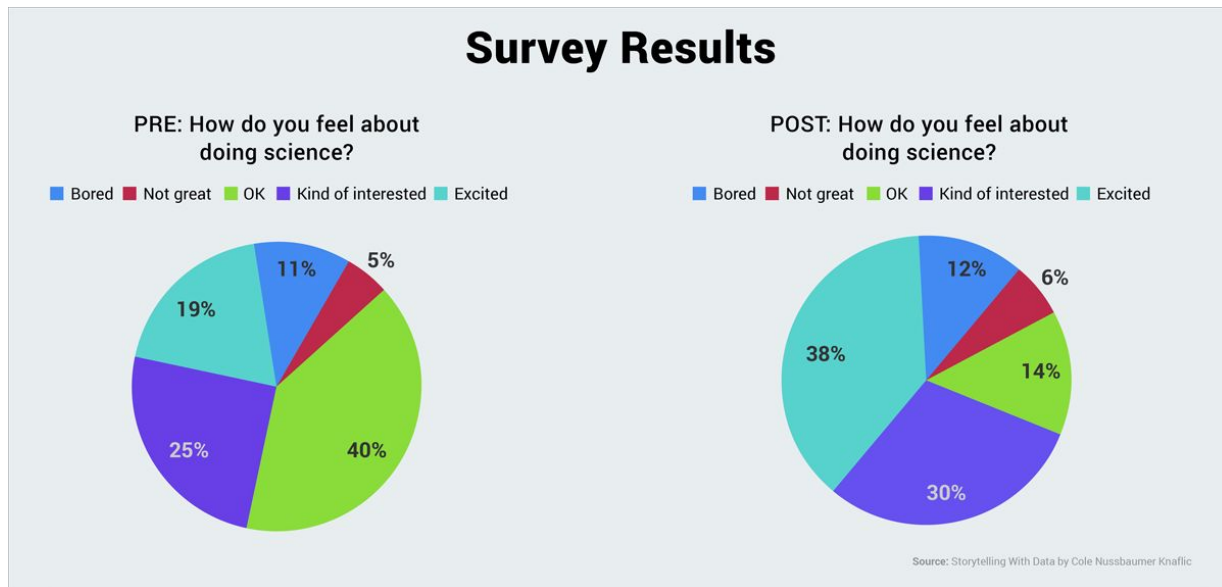
Retail price over time by product



Source: Storytelling With Data by Cole Nussbaumer Knaflic

Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing grouping the differences

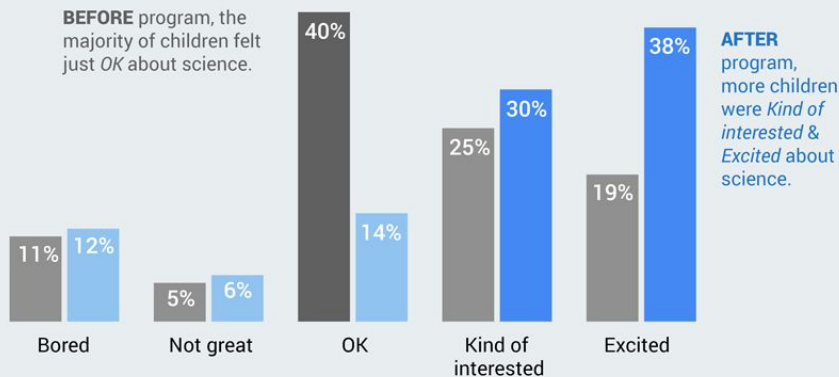


Choosing the Right Chart to Emphasize the Message

Making the case by emphasizing grouping the differences

Pilot program was a success

How do you feel about science?



Source: Storytelling With Data by Cole Nussbaumer Knaflic

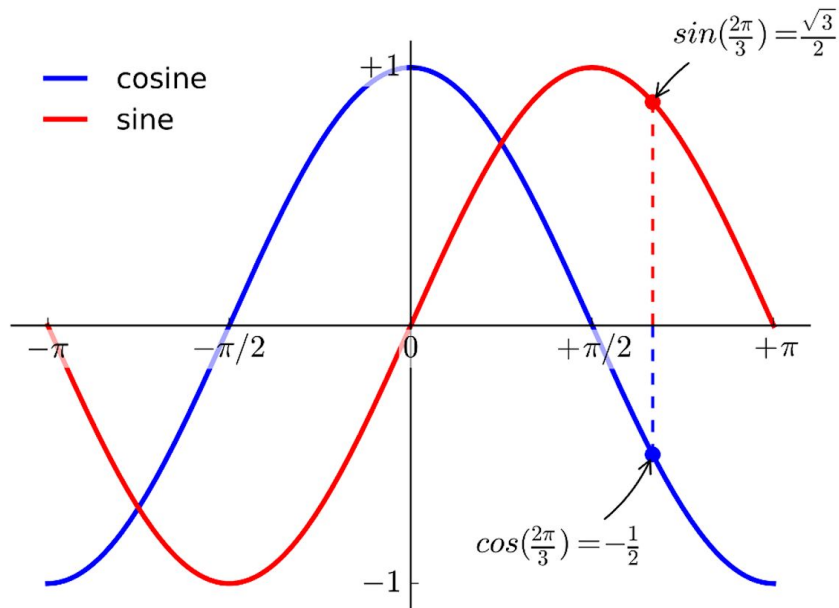
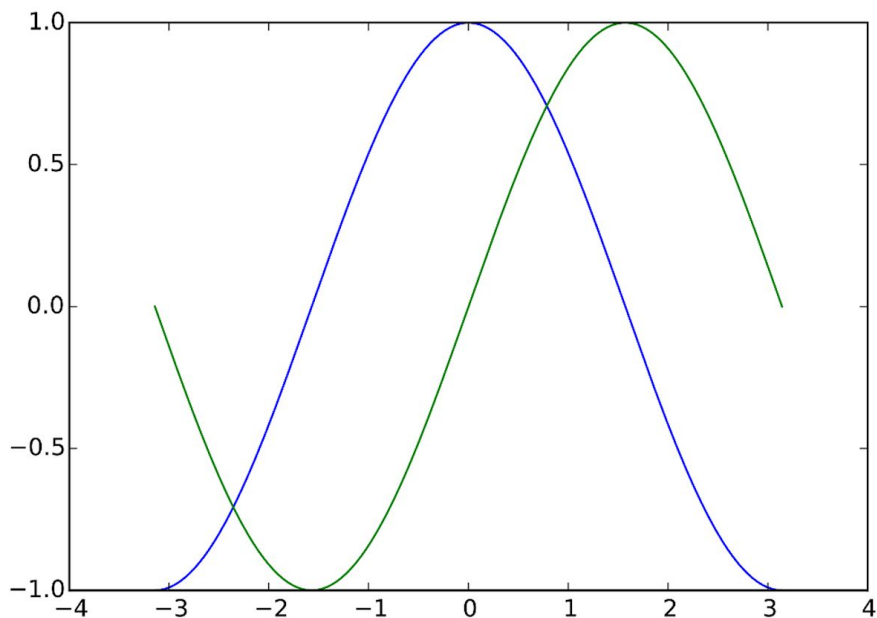
2. Aesthetics

It's not just about looking pretty;
it's about being easily
interpretable.

Don't trust the default settings

Don't trust the default settings

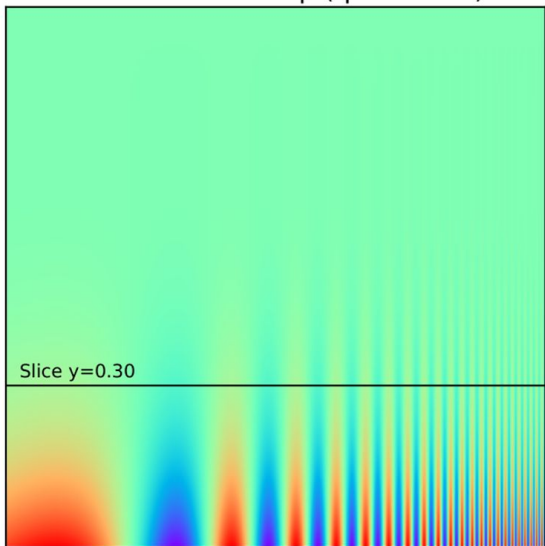
Make it as easy as possible to understand your figure



Don't trust the default settings

Be mindful of how you use color!

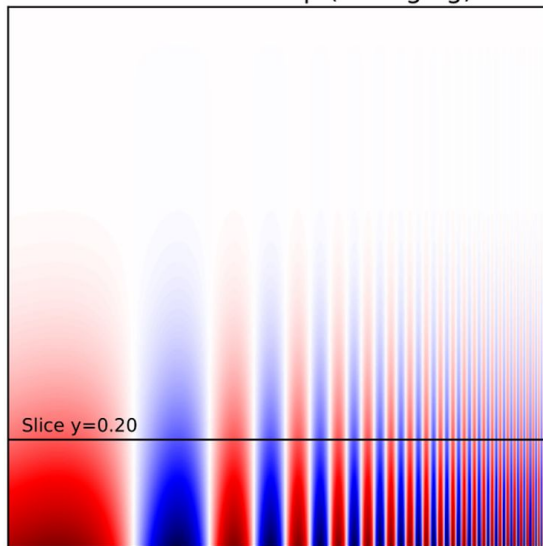
Rainbow colormap (qualitative)



Slice detail



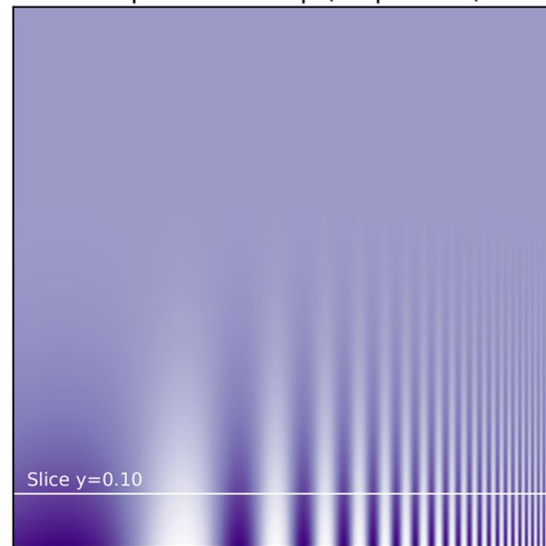
Seismic colormap (diverging)



Slice detail



Purples colormap (sequential)

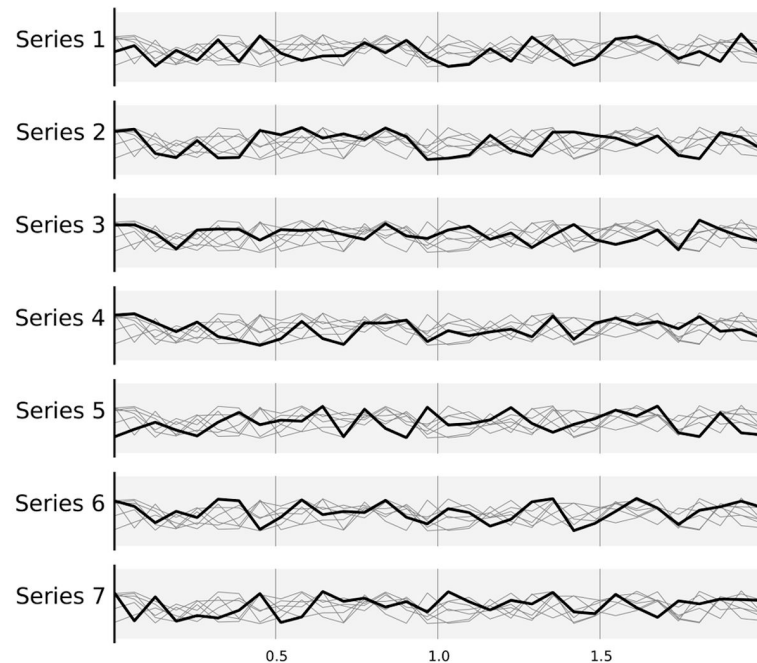
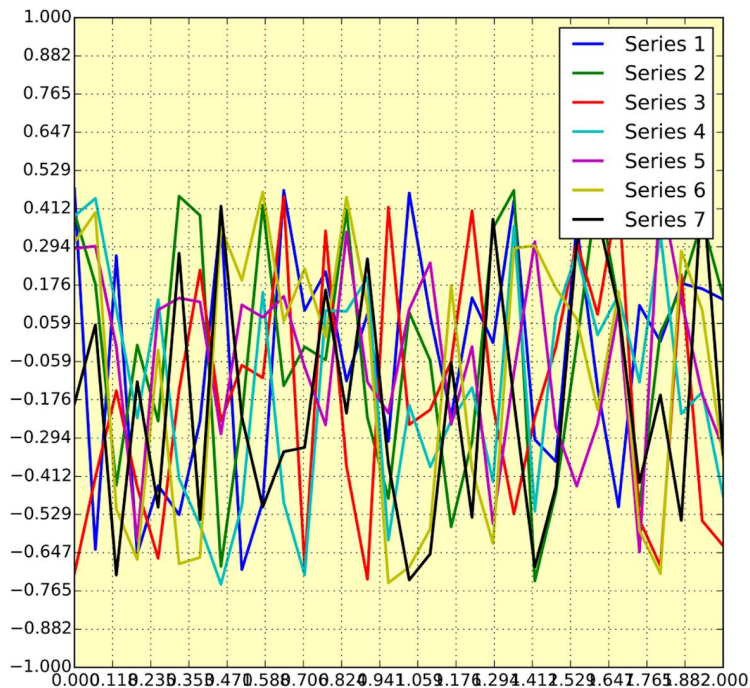


Slice detail



Don't trust the default settings

Avoid chartjunk



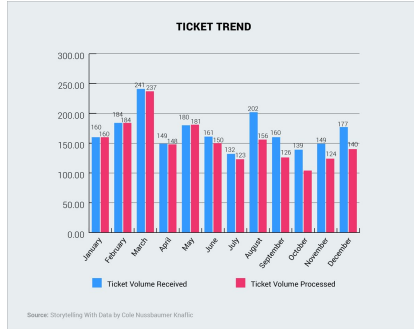
3. Communicate Clearly

How to improve the defaults

1. Know your audience & message
2. **Label axes, include title & legend in a legible size & font.**
3. Show control (baseline) data first and in the “neutral” (gray) color
4. X-axis should be the independent variable (time, category)
5. Scale things appropriately [try to avoid clipping the Y-axis...]

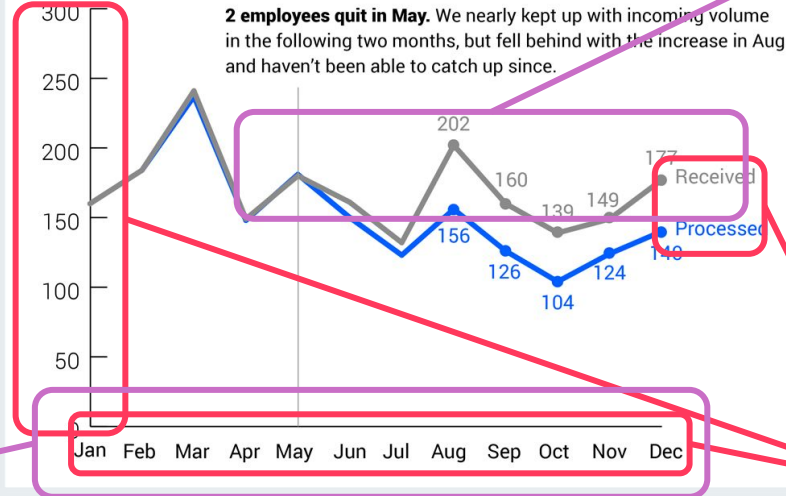
Application of Principles

1. Message & Audience



Please approve the hire of 2 FTEs
to backfill those who quit in the past year

Ticket volume over time



3. Baseline neutral

2. Labelled & Legible

4. Independent Variable

Application of Principles

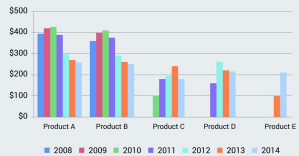
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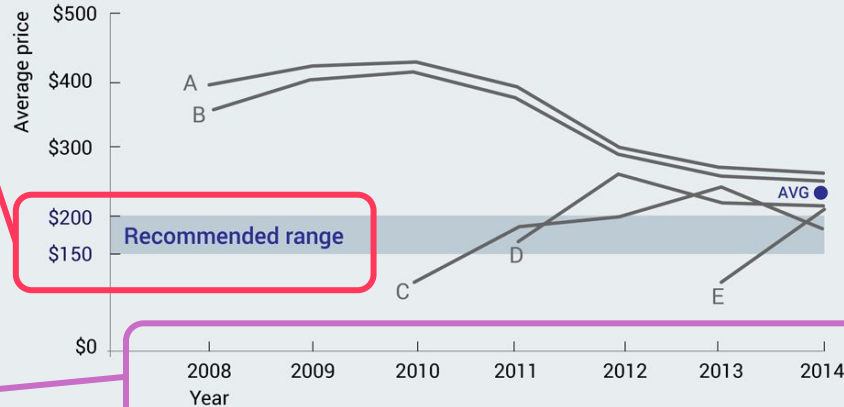
2. Labelled & Legible

3. Baseline neutral

Average Retail Product Price per Year



Retail price over time by product



4. Independent Variable

Applications of Principles

1. Message & Audience

2. Labelled & Legible

3. Baseline neutral

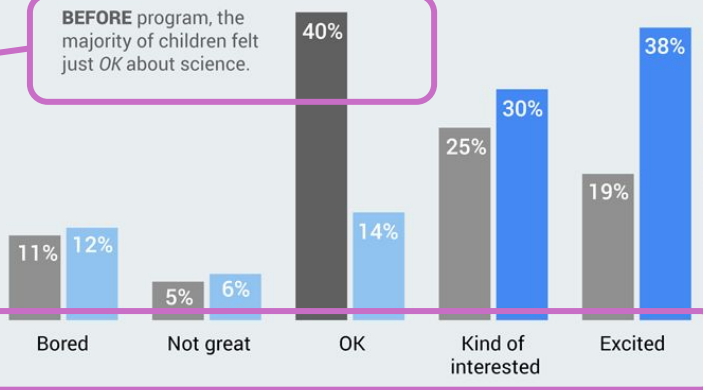
4. Independent Variable

Pilot program was a success

How do you feel about science?

BEFORE program, the majority of children felt just OK about science.

AFTER program, more children were Kind of interested & Excited about science.

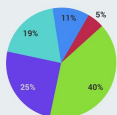


Source: Storytelling With Data by Cole Nussbaumer Knaflic

Survey Results

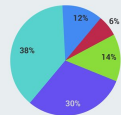
PRE: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



POST: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



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Label & Make Legible